

CSE/MB620 · DEEP LEARNING WITH PYTORCH · PROJECT VIDEO

Neural Atlas
神经图谱

A 3D atlas of 21 neural-network architectures, with an AI core that swaps in the best-fitting network to reason about your question.

[Your name] · [Student ID] · October 2026

THE IDEA · 项目动机

Choosing an architecture is choosing how a problem gets taken apart.

01 · SEE

21 architectures in one space

Perceptron to DiT, each with its real layer shapes and a runnable PyTorch script.

21 种网络放进同一个 3D 空间

02 · REASON

Route a question to a network

The AI core uses the chosen network’ s layer order as a reasoning template, then compares it with a single-pass answer.

按问题换脑，和单次直答对比

03 · TRAIN

Bring your own data

Images, audio or tables: train in the browser, compare with the online model, deploy only when better.

载入数据 → 训练 → 对比 → 上线

SYSTEM OVERVIEW · 系统结构

Three parts: the page, the model, the code

BROWSER PAGE

three.js 3D atlas

- 21 models on one ring, code screens above
- Architecture inspector + bloom pass
- Data bay: features, MLP training, Adam
- Chinese / English / both, everywhere

CLAUDE · 5 CALLS / QUESTION

Reasoning layer

- Route → structured JSON plan
- A: single pass (quick tier)
- B: conditioned on the plan (default tier)
- Cross-domain lenses + a judge

PYTORCH · D:\neural-forge

Model zoo + backend

- 21 scripts in models/zoo
- train / predict / selftest per model
- Shared common.py, _data, _runner
- API key only in .env, read by server.py

The page embeds every script’s full source, so what you see in 3D and what you can run locally are the same models. 页面里的 3D 结构和本地可运行的 PyTorch 代码是同一套模型。

THE 3D ATLAS · 主舞台

21 networks around one AI core

Tier 1 • the models

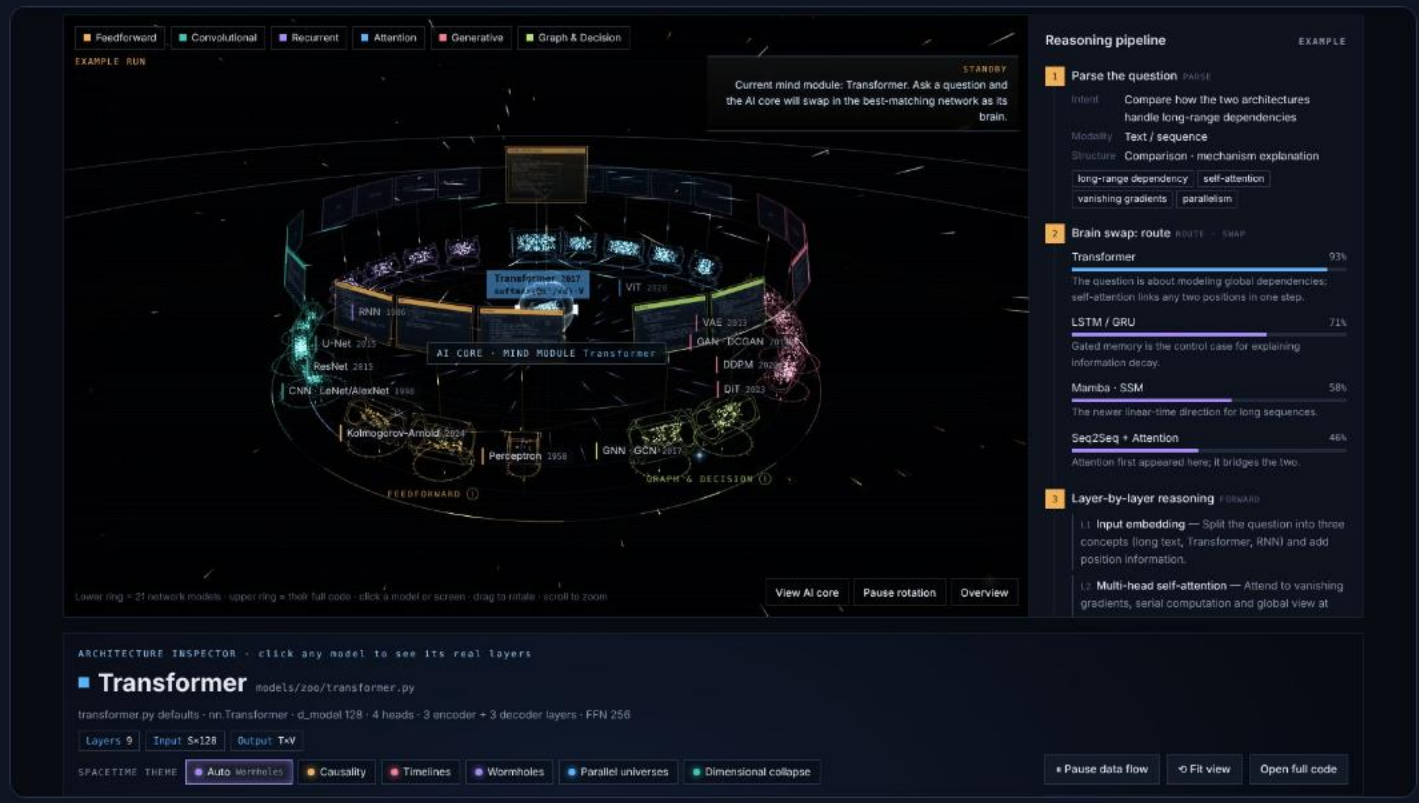
Six families on one ring: feedforward, convolutional, recurrent, attention, generative, graph & decision.

Tier 2 • the code

Each model’ s own source screen floats above it; click to open the full file.

Centre • the router

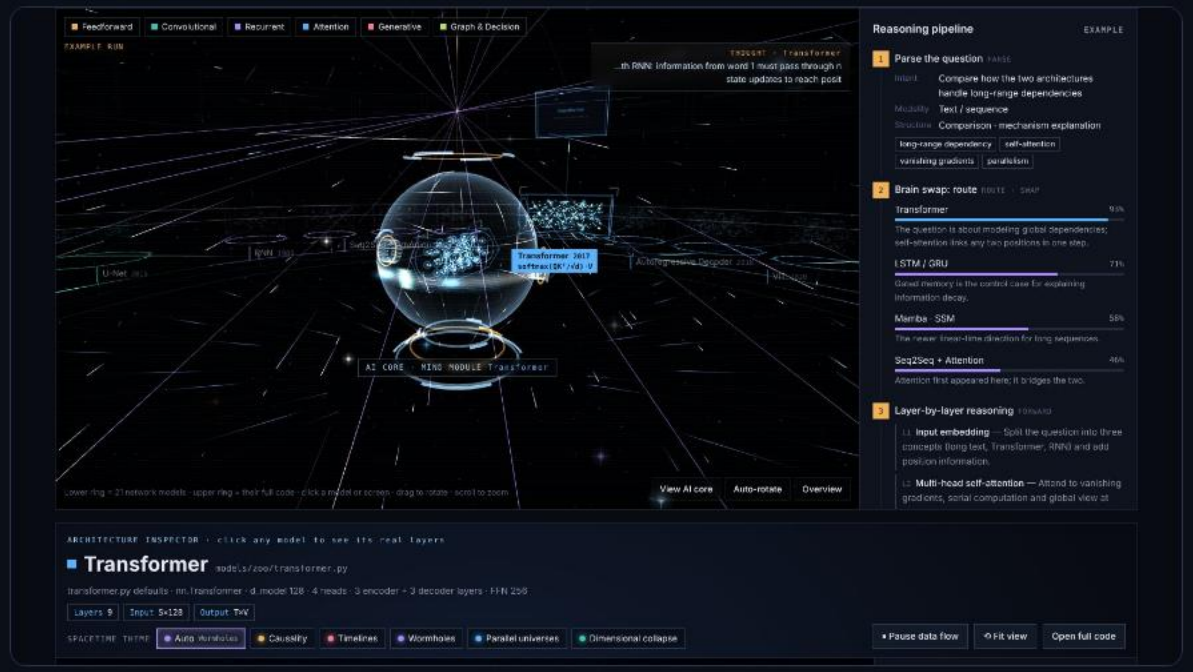
The skull in the middle holds the current “mind module”; rays link it to every model.



THE REASONING PIPELINE · 换脑推理

Route, swap, reason, refine, answer

① **Route** top 4 of 21 → ② **Swap** new brain → ③ **Forward ×5** layer order → ④ **Refine ×3** flaw + fix → ⑤ **Generate** + ⑥ judge



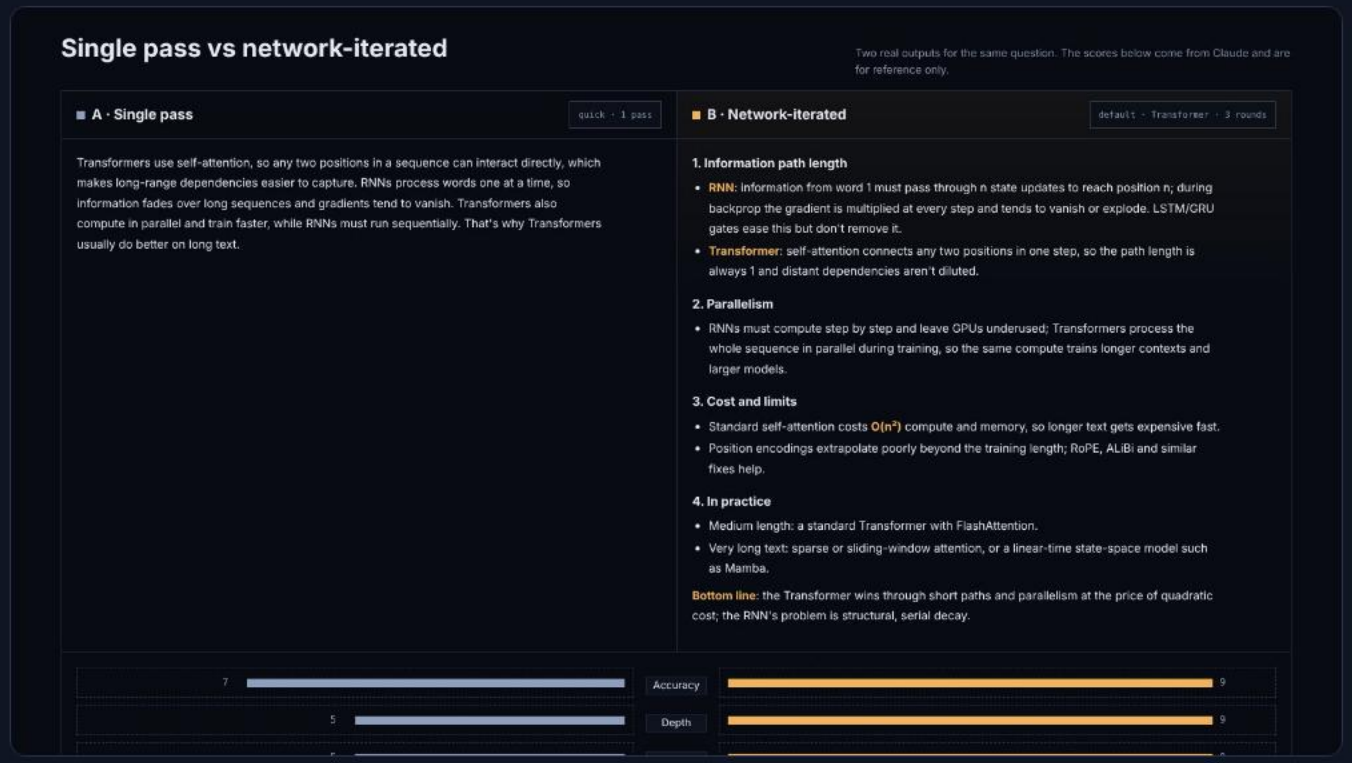

```
answer = Generate( q, Plan(q),
Refine³(Plan(q)) )
```

The routing call returns only JSON — intent, modality, scores, 5 layer steps, 3 critiques — so every animation in 3D comes straight from that plan.

路由调用只输出结构化 JSON，3D 中每一步动画都来自这份计划。

SIDE-BY-SIDE RESULT · 结果对比

Single pass vs network-iterated



	A · single	B · iterated
Calls	1	3 + 1
Tier	quick	default
Plan	none	routed JSON
Self-check	none	3 rounds

The page measures real time-to-first-token, total time and length for both, and a judge call scores accuracy, depth, structure and actionability.

真实计时 + 评审打分，双语模式下两种回答都分中英两段。

CROSS-DOMAIN LENSES · 跨学科视角

Other disciplines look at the same question

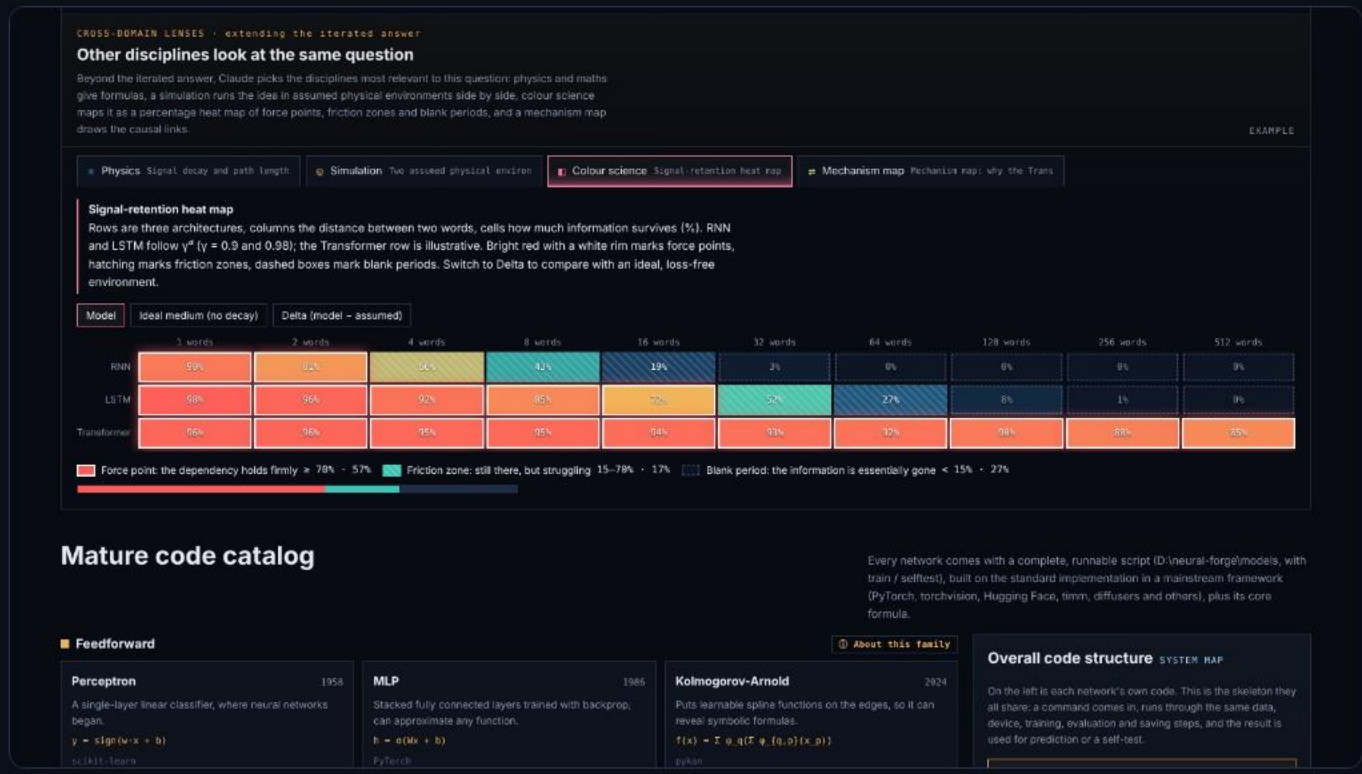
Formula — Physics or math equations, each with its meaning

Simulation — A 2D spring sandbox run in two assumed environments

Heat map — Colour by percentage: blank period, friction zone, force point — vs a baseline

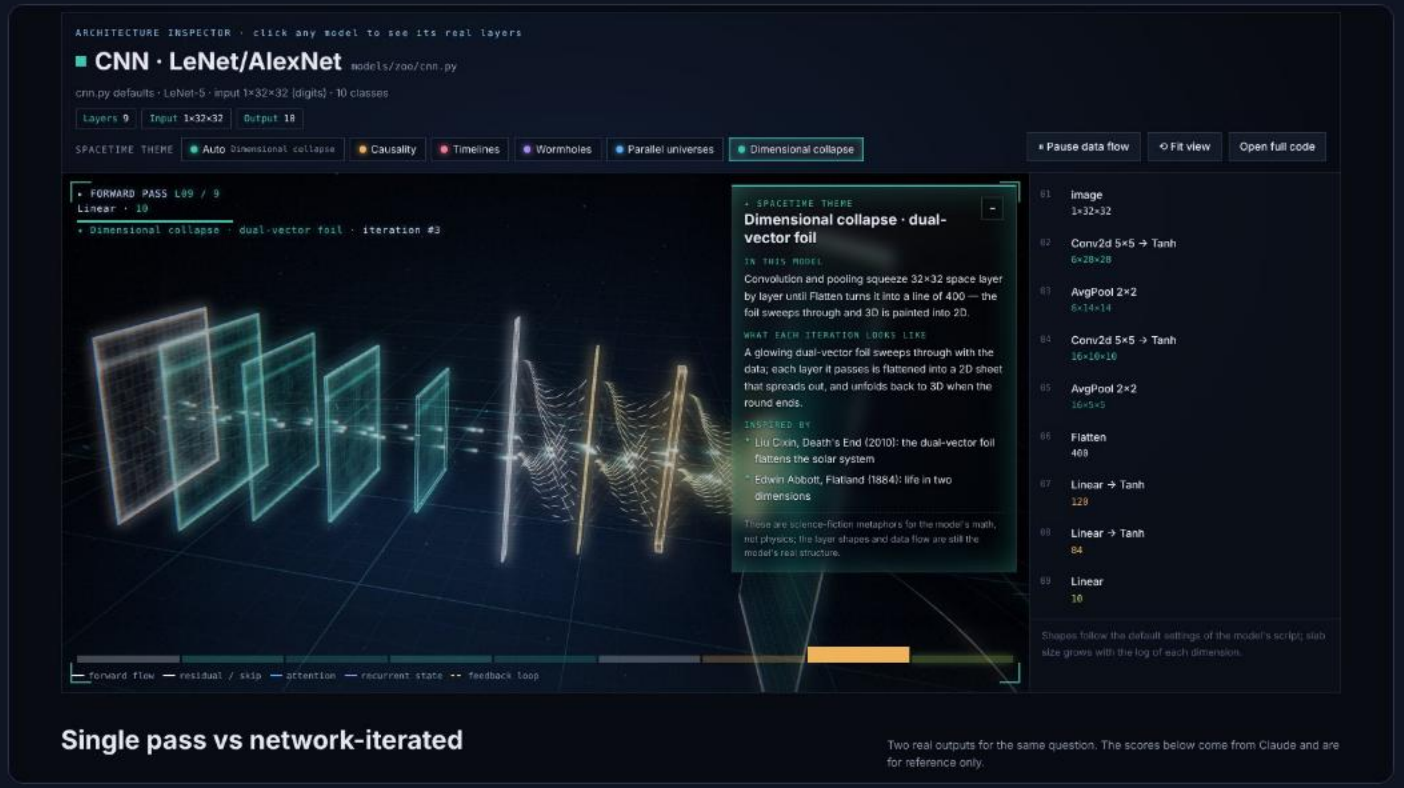
Graph — A mechanism map of what leads to what

公式、模拟、颜色学热图（空白期 / 磨合区 / 用力点）和机制图



ARCHITECTURE INSPECTOR · 结构透视

Real layer shapes from each script’ s defaults



cnn.py · LeNet-5 defaults

Layer	Shape
Input	1×32×32
Conv 5×5	6×28×28
AvgPool	6×14×14
Conv 5×5	16×10×10
AvgPool	16×5×5
Flatten	400
Linear ×3	120 → 84 → 10

点任意模型，下方显示真实层结构；板块大小随维度的对数增长。

SPACETIME THEMES · 时空主题

Sci-fi metaphors tied to each model’s math



Wormholes

Self-attention and skips join distant positions in one step.



Light cones

GPT’s causal mask: each token sees only the past.



Parallel universes

MoE routes each token to its top-2 of 4 experts.



Time loops

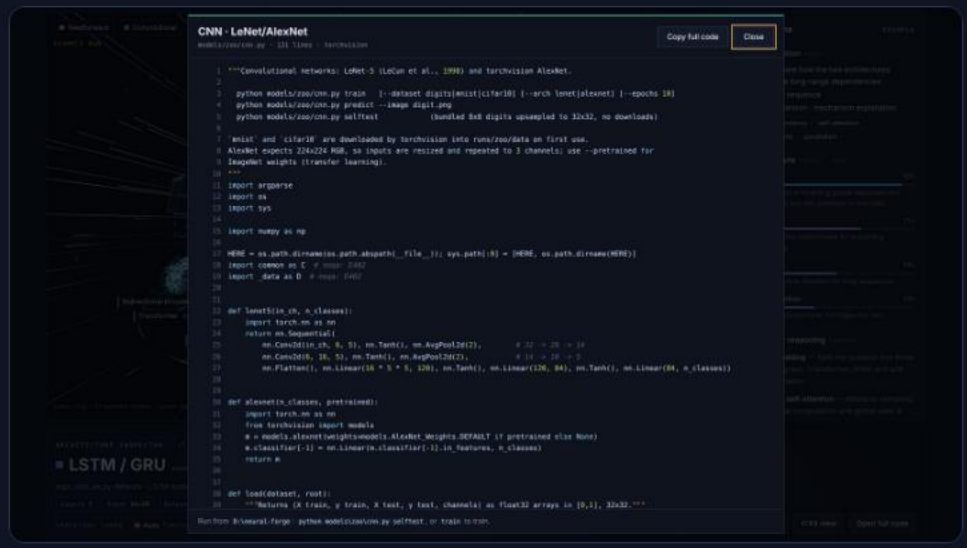
LSTM carries hidden state round the loop; the forget gate chooses what survives.

Five themes (also: dimensional collapse for CNN/ViT). Each is a metaphor for the math, never physics — the layer shapes and data flow stay the model’s real structure. 科幻只是比喻，层结构仍然真实。

PYTORCH MODEL ZOO · 成熟代码目录

Every glyph is a runnable script

Script	Default training command (options)
cnn.py	train --dataset digits|mnist|cifar10 --arch lenet|alexnet
vit.py	train --data <folder> (one sub-folder per class)
rnn.py	train --csv series.csv --column value
gpt.py	train --text my_corpus.txt (fine-tune GPT-2)
gnn.py	train --dataset karate|cora|citeseer
dqn.py	train --env CartPole-v1



21 scripts share common.py, _data and _runner; each has train, predict and selftest.

21 个脚本共用公共模块，均支持 train / predict / selftest。

DATA BAY · 数据舱

Load images, audio or a table — see every step

Image folder

Sub-folder = label. 12×12 grey pixels + RGB histogram + brightness, contrast, edges → 174 features.

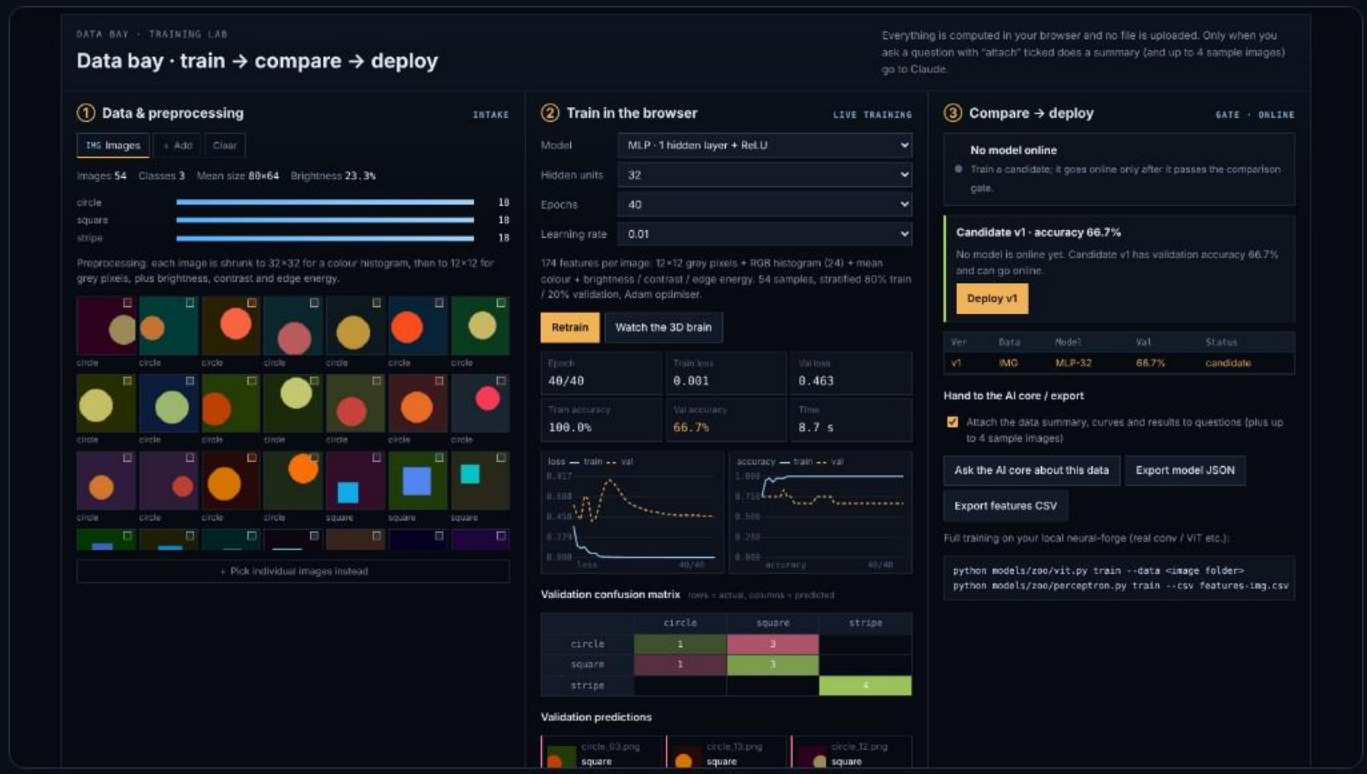
Audio clips

Mono, ~16 kHz, 512-point FFT → 20 log bands; mean/std + RMS, zero-crossing, centroid → 46 features.

Data table

CSV / JSON / XLSX. Column profiles; pick a target; classification or regression detected.

全部在浏览器里计算，文件不上传。



TRAIN → COMPARE → DEPLOY · 训练 → 对比 → 上线

A new model goes online only if it beats the current one.

Every training run makes a candidate version. The gate compares its validation score with the online version on the same data and task; worse candidates stay offline unless forced.

Test run (synthetic data)	Model	Validation
54 images, 3 classes · v1	MLP-32	66.7% · online
Same images · v2	Perceptron	83.3% · passes gate
Iris-style table, 120 rows	MLP-32	100% accuracy



After deploy: drop new files for online inference (top-3 probabilities); export model JSON or features CSV for full PyTorch training.

EXPERIENCE DETAILS · 体验细节

Built to be read, in two languages



Family wiki

Click a family name: a translucent card explains what it processes and its principle.

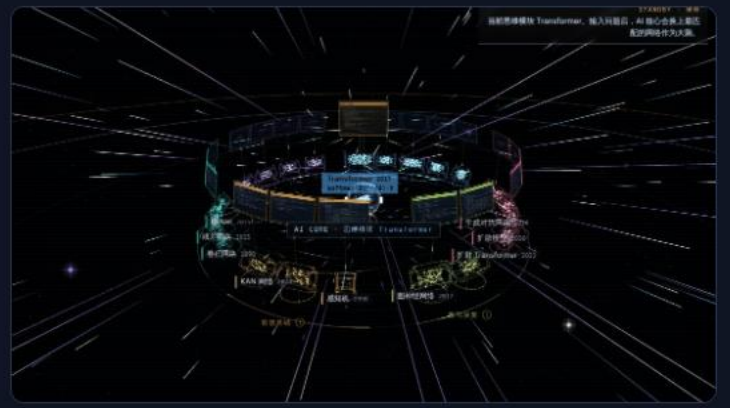
中文 · EN · 中英

中文
注意力一步连接任意两个位置…

English
Attention links any two positions…

Three language modes

UI, 3D labels, prompts and model output all switch; “both” returns paired sections.



Warp background

Star streaks change colour and length with speed; a 13 s cycle ends in a space collapse.

ENGINEERING & TESTING · 工程与测试

How it was built and checked

Implementation

- three.js r128, custom shaders, two WebGL renderers, own bloom pass
- Backprop and Adam written by hand in JS; radix-2 FFT for audio
- Claude via the page’ s sample API: JSON mode, streaming, stop
- All 21 scripts embedded as source data in one page

Verification

- Headless Chromium renders of every view with real three.js
- Mock Claude to test all 5 prompts in each language mode
- Synthetic image, audio and table sets for the data bay
- Fixed: inspector blank when built at zero size → camera guard + self-heal

Security: the Gemini key for image generation lives only in .env on the local backend and never reaches the browser. 密钥只放在本地 .env。

LIMITS AND NEXT STEPS · 局限与下一步

From a visual atlas to a working tool

Honest limits today

- Browser training uses hand-made features, not real convolutions
- “Brain swap” is a reasoning template, not running the network on the question
- Each live question spends 5 model calls
- Test numbers come from small synthetic data

Next

- Send data-bay datasets to the local PyTorch backend for real CNN / ViT training
- Same compare-then-deploy gate for backend models
- First real tasks: code vulnerability detection, log anomalies, medical / pet images, tabular prediction

Thank you · 谢谢